

JAEHYEOK BAE

☎ (+1) 650-512-4770 ✉ jhbae110@stanford.edu 🏠 jaehyeokbae.me

EDUCATION

Stanford University

Ph.D. in Electrical Engineering

Sep 2024 - Present

Stanford, CA

Seoul National University

B.S. in Electrical and Computer Engineering

Mar 2016 - Aug 2023

Seoul, South Korea

- Graduate Summa cum laude (5th place / 148), Total GPA : 4.23 / 4.30 (Major GPA : 4.27 / 4.30)
- * Served mandatory military service between 2018 - 2019

PUBLICATIONS (SELECTED)

- [1] **J. Bae**, Z. Shah, C. Alkan, S. Vasanaawala, J. M. Pauly, and K. Setsompop, “AdaSamp: Towards simple, subject-specific Adaptive Sampling for 3D Accelerated MRI,” in *Proceedings of the ISMRM Annual Meeting*, 2026 (**Summa Cum Laude**)
- [2] **J. Bae**, C. Alkan, S. Vasanaawala, J. M. Pauly, and K. Setsompop, “QuickSamp: Towards simple, real-time-optimized Sampling Patterns for 3D Accelerated MRI,” in *Proceedings of the ISMRM Annual Meeting*, 2025 (**Magna Cum Laude**)
- [3] K. Kyeonghun and **J. Bae** et al., “3D-LLDM: Label-Guided 3D Latent Diffusion Model for Improving High-Resolution Synthetic MR Imaging in Hepatic Structure Segmentation,” in *IEEE International Symposium on Biomedical Imaging*, 2026
- [4] A. Nurdinova, W. Huang, D. R. Abraham, **J. Bae**, Y. Lin, K. Setsompop, and B. A. Hargreaves, “Neural space-time modeling for motion-corrected MR reconstruction,” in *MICCAI Workshop on Machine Learning for Medical Image Reconstruction*, 2025
- [5] S. Hong, **J. Bae**, J. Lee, and S. Y. Chun, “Adaptive Selection of Sampling-Reconstruction in Fourier Compressed Sensing,” in *Proceedings of The European Conference on Computer Vision (ECCV)*, 2024
- [6] **J. Bae**, J. Lee, and S. Kim, “PNI : Industrial Anomaly Detection using Position and Neighborhood Information,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023
- [7] J. Kim, **J. Bae**, G. Park, D. Zhang, and Y. Kim, “N-ImageNet: Towards Robust, Fine-Grained Object Recognition with Event Cameras,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021

EXPERIENCE

Setsompop Lab.

Ph.D. Candidate (Advisors: Prof. Kawin Setsompop and Prof. John M. Pauly)

Sep 2024 - Present

Lab. for Imaging Science and Technology

Research Intern (Advisor: Prof. Jongho Lee)

Mar 2023 - Nov 2023

Gauss Labs Inc.

Computer Vision Applied Scientist Intern

Jan 2022 - Nov 2022

- Proposed a novel anomaly detection and localization algorithm for industrial datasets, by training a normal feature distribution using position and neighborhood information of local features.

3D Vision Lab.

Research Intern (Advisor: Prof. Youngmin Kim)

April 2020 - Mar 2021

SKILLS & LANGUAGES

Programming Languages	Python, MATLAB, C/C++
Libraries/Frameworks	PyTorch, TensorFlow, Scikit-learn
Languages	Korean (Native), English

SCHOLARSHIPS

ILJU Overseas Ph.D. Scholarship , <i>ILJU Academy & Culture Foundation</i>	2024 - Present
- 5-year Research award and merit-based prize for academic excellence.	
Presidential Science Scholarship , <i>Korea Student Aid Foundation (KOSAF)</i>	2016 - 2021
- 4-year Full tuition and additional stipends (\$5,000 / year) for academic excellence.	

HONORS AND AWARDS

Summa Cum Laude Merit Award , <i>ISMRM Annual Meeting</i>	2026
Magna Cum Laude Merit Award , <i>ISMRM Annual Meeting</i>	2025
2nd place in the 2022 SNU FastMRI Challenge, <i>Seoul National University</i>	2022

TEACHING AND VOLUNTEERING EXPERIENCES

Contest Coordinator	
Organized and supervised the 2023 SNU FastMRI Challenge, <i>Seoul National University</i>	2023
Undergraduate Student Tutor , <i>Seoul National University</i>	2017 - 2023
Subjects: Random Variables, Communications, Algorithms, Foundation of Physics 1 & 2.	
Mentoring	
International Students Integrated Peer Tutoring Program, <i>Seoul National University</i>	Fall 2021
University Student's Talent Donation Camp, <i>Korea Student Aid Foundation</i>	2016, 2017

EXTRACURRICULAR ACTIVITIES

SNU Tomorrow's Edge Membership(STEM)	Mar 2021 - Aug 2022
<i>12th Vice Chairman of SNU Honor Society</i>	<i>Seoul National University</i>
· A student organization that presents a vision of "Outstanding engineering knowledge, broad consideration of different perspectives, and active communication with society." · Served as a leader of 200+ STEM members, encouraged activities; interdisciplinary academic seminars, and mentoring programs for teenagers.	
Republic of Korea Army	Jan 2018 - Sep 2019
<i>Soldier of Information Security Specialist</i>	<i>Gyeryong, Republic of Korea</i>
· Completed mandatory military service (honorable discharge) · Information protection tasks to prevent computer viruses and hacking at the Army headquarters.	